Experiment 1

Frequency Analysis and Probability in Data Science Using Python

Subject: Data Science Lab | Dataset: Titanic | Mapped CO(s): CO1, CO2

1. Aim

To perform frequency analysis, calculate probability measures, and analyze correlation using the Titanic dataset in Python.

2. Key Concepts & Definitions

Frequency Analysis

A statistical method to count how often each category or value appears in a dataset.

$$f_i = \text{count of category } i$$

Relative Frequency

The proportion (percentage) of observations in each category, where n = total observations.

$$\text{Relative Frequency (\%)} = \frac{f_i}{n} \times 100$$

Cumulative Frequency

Running total of frequencies up to a given category.

$$CF_k = \sum_{i=1}^{k} f_i$$

Probability P(A)

Measure of how likely an event is to occur, ranging from 0 (impossible) to 1 (certain).

$$P(A) = \frac{n(A)}{n}$$

Event

A subset of outcomes from a sample space. Example: surviving the Titanic disaster.

Sample Space (S)

The set of all possible outcomes. Example: S = {Survived, Did Not Survive}.

Contingency Table

A two-way table showing the frequency distribution of two categorical variables together.

Joint Probability

Probability that two events A and B occur together.

$$P(A \cap B) = \frac{n(A \cap B)}{n}$$

Marginal Probability

Probability of a single event, ignoring other variables.

$$P(A) = \frac{n(A)}{n}$$

Conditional Probability

Probability of event A given that event B has already occurred.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}$$

Variance Var(X)

Measures how spread out a single variable is from its mean.

$$\text{Var}(X) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Standard Deviation (sigma)

Square root of variance. Same units as the original data.

$$\sigma = \sqrt{\text{Var}(X)}$$

Covariance Cov(X, Y)

Measures the direction of linear relationship between two variables.

$$\text{Cov}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Interpretation: Cov(X,Y) > 0: variables increase together; Cov(X,Y) < 0: one increases as other decreases; Cov(X,Y) ~ 0: no linear trend.

Correlation (r)

Standardized covariance. Measures strength and direction of linear relationship (-1 to +1).

$$r = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

Covariance vs Correlation

Covariance depends on units (unbounded). Correlation is unitless and always in [-1, +1].

Pearson Correlation Coefficient

Most common measure of linear correlation between two numerical variables.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

3. Symbol Glossary

Symbol	Meaning
n	Total number of observations
f_i	Frequency of category i
x_i, y_i	i-th values of variables X and Y
x-bar, y-bar	Sample means of X and Y
P(A)	Probability of event A
P(A cap B)	Joint probability of A and B
P(A | B)	Conditional probability of A given B
Var(X)	Variance of X
sigma	Standard deviation
Cov(X, Y)	Covariance between X and Y
r	Pearson correlation coefficient

4. Important Formulae (Summary)

(A) Mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

(B) Absolute, Relative & Cumulative Frequency

$$f_i = \text{count of category } i$$

$$\text{Relative Frequency (\%)} = \frac{f_i}{n} \times 100$$

$$CF_k = \sum_{i=1}^{k} f_i$$

(C) Probability Measures

$$P(A \cap B) = \frac{n(A \cap B)}{n}$$

$$P(A) = \frac{n(A)}{n}$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}$$

(D) Variance & Standard Deviation

$$\text{Var}(X) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$\sigma = \sqrt{\text{Var}(X)}$$

(E) Covariance & Correlation

$$\text{Cov}(X, Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

(F) Range of Pearson r

$$-1 \leq r \leq +1$$

- r = +1 -> perfect positive correlation
- r = -1 -> perfect negative correlation
- r = 0 -> no linear relationship

5. Algorithm

- 1. Import Pandas, NumPy, Matplotlib, Seaborn
- 2. Load Titanic dataset
- 3. Create frequency table for passenger class
- 4. Calculate absolute, relative, and cumulative frequencies
- 5. Create contingency table: Sex vs Survived using pd.crosstab()
- 6. Calculate joint, marginal, and conditional probabilities
- 7. Select Age and Fare; handle missing values
- 8. Compute Pearson correlation
- 9. Plot heatmap, scatter plot, and stacked bar chart

6. Experimental Results

Frequency Table - Passenger Class

Class	Absolute Freq	Relative (%)	Cumulative
First	216	24.24	216
Second	184	20.65	400
Third	491	55.11	891

Contingency Table - Sex vs Survived

Sex	Not Survived (0)	Survived (1)	Total
female	81	233	314
male	468	109	577
All	549	342	891

Joint Probability P(Sex and Survived)

Sex	0	1
female	0.0909	0.2615
male	0.5253	0.1223

Conditional Probability P(Survived | Sex)

Sex	Not Survived	Survived
female	0.2580	0.7420
male	0.8111	0.1889

Pearson Correlation - Age vs Fare

- r = 0.0961 -> very weak positive linear relationship

7. Exercise Answers

- Frequency table for Embarked: absolute, relative, cumulative frequencies computed
- P(Survived) = 0.3838 (38.38%)
- Pearson r (Age, Fare) = 0.0961
- Stacked bar chart: survival by passenger class plotted

8. Viva Q&A

Q: What is covariance?

A: Measure of how two variables change together (see formula above).

Q: What is correlation?

A: Standardized covariance: r = Cov(X,Y)/(sigma_X sigma_Y), range -1 to +1.

Q: Difference between covariance and correlation?

A: Covariance is unbounded and unit-dependent; correlation is standardized.

Q: What is conditional probability?

A: P(A|B) = P(A and B)/P(B) - probability of A given B occurred.

Q: What does r = 0.0961 mean?

A: Very weak positive linear correlation between Age and Fare.

9. Conclusion

Frequency analysis, probability calculations, and correlation analysis were successfully performed on the Titanic dataset using Python.